

Concept 2 | Exponential Functions and Their Applications

English Student Edition • Focused formulas and guided practice

Formula opener

Exponential

$$y = ab^x$$

Growth

$$A = A_0(1 + r)^t$$

Decay

$$A = A_0(1 - r)^t$$

Classified Practice

Exponential Functions and Their Applications

Q001 Written

Draw the graphs of $y = 3^x$ and $y = (1/3)^x$, using the points $x = -1, 0, 1$.

Q002 Written**Draw $y = 2^x$ and $y = (1/2)^x$.**

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Q003 Written

Draw the six source graphs $y = 4^x$, $y = 10^x$, $y = (2/3)^x$, $y = (1/4)^x$, $y = (1/10)^x$, $y = (3/2)^x$.

Q004 Written**Draw $y = \pi^x$ for $-1 \leq x \leq 2$, marking the key points.**

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Q005 Written

Order the source sets 0.7^{-3} , 0.7^{-1} , 0.7^2 and $5/8$, $5/64$, $6/16$ using inequality signs.

Q006 Written

Order the three displayed Try sets: $5^0, 5^{-4}, 5^{3/2}$; $(1/3)^7, (1/3)^2, (1/3)^{-5}$; **and** $3^0, 3^2, 3^{-2}$.

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Exponential Functions and Their Applications

Q007 Written

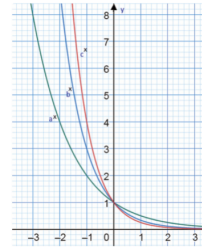
Order the source sets with bases 3, 2, $3/2$, 7, 0.9, and 0.1, including the radical/fraction forms shown in the lesson.

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Q008 **Written**

Order $(\sqrt{2})^5$, $(\sqrt[3]{4})^3$, $(\sqrt[4]{8})^2$, then use the opposite graph to compare the labelled values a, b, c .



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Exponential Functions and Their Applications

Q009 Written

Solve the six Warm Up items: $4^{x-1} = 8^{x+3}$, $3^x = 81$, $7^x = 7$, $2^{2x-1} \geq 1/2$, $(1/27)^x < (1/3)^{x+4}$, and $8 \geq 2^{2x-1}$.

Q010 Written

Solve the eight Try equations/inequalities displayed in lesson 6-7, including $9^x = 243$, $3^{2x+1} = 125$, $2^{3x} \leq 9^{x+1}$, and the reciprocal-base inequalities.

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Q011 Written

Solve all 24 source equations and inequalities in Exercise (a)–(x), including the 2, 3, 4, 5, 8, 9, 25, 27, 32, 64, 81, 125 common-base forms.

Q012 Written

A food-safety model requires $4^{x^2-x} > 16$. Find when the batch becomes unsafe.

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Q013 Written

Solve $4^x - 3 \cdot 2^x - 16 = 0$ and $8 \cdot 3^{-2x+1} - 3^{-x} \leq 0$ by substitution.

Q014 Written

Solve the four Try items in lesson 6-8: two quadratic equations and two inequalities in powers of 2 and 3.

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Q015 **Written****Solve the twelve source items (a)–(l), each reducible to a quadratic in 2^x or 3^x .**

Q016 Written

Solve completely for real x : $(1/4)^{x^2-2x} - 5(1/2)^{x^2-2x} + 4 \leq 0$.

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Q017 Written

For $-1 \leq x \leq 3$, find the maximum and minimum of $y = 2^{2x} - 4 \cdot 2^x$, with the corresponding x -values.

Q018 Written

For $-1 \leq x \leq 2$, find the maximum and minimum of $y = -9 + 2 \cdot 3^x + 3^{2x}$, with the corresponding x -values.

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Q019 Written

Solve the three source extrema models: $-3 \leq x \leq 2$, $y = 2 \cdot 9^x - 12 \cdot 3^x + 7$; $-1 \leq x \leq 3$, $y = -4 + 2^{2x} - 3^{x+2}$; and the third displayed interval/model.

Q020 Written**For $-1 \leq x \leq 2$, optimise $y = 2 \cdot 4^x - 2^{x+2} + 11$.**

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Q021 Written

For $f(x) = a^x$, $g(x) = (1/a)^x$ with $a > 1$, analyse $h(x) = f(x) + g(x)$.

Q022 MCQ

Which base gives a strictly decreasing exponential graph?**Choose the correct option.**

A 2

B 3

C $\frac{1}{2}$ D π

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Q023 MCQ

The horizontal asymptote of $y = a^x$ is:**Choose the correct option.**

A $x = 0$

B $y = 0$

C $y = 1$

D $x = 1$

Q024 MCQ

The range of $y = 5^x$ is:**Choose the correct option.**A $\mathbb{R}$ B $[0, \infty)$ C $(0, \infty)$ D $(-\infty, 0)$

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Q025 MCQ

For $y = \pi^x$ on $[-1, 2]$, which point is on the graph?

Choose the correct option.

A $(0, \pi)$

B $(1, \pi)$

C $(2, \pi)$

D $(-1, \pi)$

Q026 MCQ**For $0 < a < 1$ and $r < s$, which is true?****Choose the correct option.**

A $a^r < a^s$

B $a^r > a^s$

C $a^r = a^s$

D Cannot compare

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Q027 MCQ

The first step when bases differ is to:**Choose the correct option.**

- A add exponents
- B make the bases equal
- C take logarithms immediately
- D ignore the base

Q028 MCQ

Which is largest? 3^{-2} , 3^0 , 3^2 **Choose the correct option.**A 3^{-2} B 3^0 C 3^2

D They are equal

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Q029 MCQ

The challenge term $(\sqrt[3]{4})^3$ equals:

Choose the correct option.

A $2^{3/2}$

B 2^2

C 2^3

D 4^3

Q030 MCQ**Solving $4^{x-1} = 8^{x+3}$ gives:****Choose the correct option.**

A $x = -11$

B $x = -3$

C $x = 3$

D $x = 11$

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Q031 MCQ

For $0 < a < 1$, an exponential inequality requires:**Choose the correct option.**

- A the same exponent order
- B a reversed exponent order
- C no base conversion
- D squaring both sides

Q032 MCQ

The solution of $3^x = 81$ is:**Choose the correct option.**

A 2

B 3

C 4

D 5

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Exponential Functions and Their Applications

Q033 MCQ

The food-safety condition $4^{x^2-x} > 16$ is unsafe for physical time when:

Choose the correct option.

A $0 \leq x \leq 2$

B $x > 2$

C $x < -1$

D $x < 2$

Q034 MCQ

In substitution methods, $t = a^x$ must satisfy:

Choose the correct option.

A $t < 0$

B $t = 0$

C $t > 0$

D $t \in \mathbb{R}$

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Q035 MCQ

The positive root of $t^2 - 3t - 16 = 0$ is:**Choose the correct option.**

A $(3 - \sqrt{73})/2$

B $(3 + \sqrt{73})/2$

C 8

D 16

Q036 MCQ

The challenge quadratic in t is:**Choose the correct option.**

A $t^2 + 5t + 4$

B $t^2 - 5t + 4$

C $t^2 - 4t + 5$

D $t^2 + 4t - 5$

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Q037 MCQ

After solving an exponential quadratic, reject roots with:**Choose the correct option.**

A $t > 1$

B $t = 1$

C $t \leq 0$

D $t < 4$

Q038 MCQ**For extrema, the first required range is the range of:****Choose the correct option.****A** x **B** $t = a^x$ **C** a **D** y only

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Q039 MCQ

The minimum of $y = t^2 - 4t$ at $t = 2$ is:**Choose the correct option.**

A -8

B -4

C 0

D 4

Q040 MCQ**The C06-K02 challenge has minimum value:****Choose the correct option.****A** 9 at $x = 0$ **B** 9 at $x = 2$ **C** 27 at $x = 0$ **D** 27 at $x = 2$

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Q041 MCQ

The global minimum of $a^x + a^{-x}$ for $a > 1$ is:**Choose the correct option.**

A 0

B 1

C 2

D It has no minimum

Q042 MCQ

At which value of x does $a^x + a^{-x}$ attain its minimum?

Choose the correct option.

A $x = -1$

B $x = 0$

C $x = 1$

D It has no stationary point

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Quiz MCQ 1 Concept Quiz · MCQ

Which statement about $y = a^x$ is always true for $a > 0$, $a \neq 1$?

Choose the correct option.

- A range $(0, \infty)$
- B range all reals
- C vertical asymptote
- D decreasing for every a

Quiz MCQ 2 Concept Quiz · MCQ

For base 0.5, increasing the exponent makes the value:

Choose the correct option.

- A larger
- B smaller
- C unchanged
- D zero

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Quiz MCQ 3 [Concept Quiz](#) · [MCQ](#)

The positive-substitution condition is:

Choose the correct option.

A $t < 0$

B $t = 0$

C $t > 0$

D t can be any real

Quiz MCQ 4 Concept Quiz · MCQ

For $y = 2(t - 1)^2 + 9$, the minimum is:

Choose the correct option.

A 0

B 1

C 9

D 11

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Exponential Functions and Their Applications

Quiz WRITTEN 5 **Concept Quiz · Written**

Sketch $y = (1/4)^x$ and state its domain, range, and asymptote.

Quiz WRITTEN 6 Concept Quiz · Written**Order** 2^{-3} , 2^0 , $2^{5/2}$.

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Exponential Functions and Their Applications

Quiz WRITTEN 7 Concept Quiz · Written

Solve $4^x - 5 \cdot 2^x + 4 = 0$.

Quiz WRITTEN 8 Concept Quiz · Written

Find the minimum of $a^x + a^{-x}$ for $a > 1$.

Answer key

Use the question number shown on each page.

- Q001** Increasing 3^x ; decreasing $(1/3)^x$; both pass through $(0, 1)$ and have $y = 0$ as asymptote.
- Q002** 2^x increases and $(1/2)^x$ decreases; domain $\mathbb{R}$, range $(0, \infty)$, horizontal asymptote $y = 0$.
- Q003** Bases 4, 10, $3/2$: strictly increasing. Bases $2/3$, $1/4$, $1/10$: strictly decreasing. Every graph has $D = \mathbb{R}$, $R = (0, \infty)$, and asymptote $y = 0$.
- Q004** Key points $(-1, 1/\pi)$, $(0, 1)$, $(1, \pi)$, $(2, \pi^2)$; the restricted graph is increasing.
- Q005** $0.7^2 < 0.7^{-1} < 0.7^{-3}$; and $5/64 < 5/8 < 6/16$.
- Q006** $5^{-4} < 5^0 < 5^{3/2}$; $(1/3)^7 < (1/3)^2 < (1/3)^{-5}$; $3^{-2} < 3^0 < 3^2$.
- Q007** Each set is ordered by converting all terms to a common base, then comparing exponents; reverse the exponent order for $0 < a < 1$.
- Q008** Rewrite every term as a power of 2: $2^{5/2}$, 2^2 , $2^{3/2}$, so $(\sqrt[4]{8})^2 < (\sqrt[3]{4})^3 < (\sqrt{2})^5$. The graph comparison follows the same increasing/decreasing rule.
- Q009** $x = -11, 4, 1, x \geq 1/4, x > 2, x \leq 2$, respectively.
- Q010** Convert each side to one base and compare exponents; the exact solution is the resulting linear equality or interval in x .
- Q011** Every item reduces to a linear condition on x ; equality items have one exact value and inequality items have the interval obtained after base/sign control.
- Q012** $x^2 - x > 2 \Rightarrow (x - 2)(x + 1) > 0$, so $x < -1$ or $x > 2$. For time $x \geq 0$, the unsafe condition is $x > 2$ hours.
- Q013** First: let $t = 2^x > 0$, $t^2 - 3t - 16 = 0$, so $t = (3 + \sqrt{73})/2$ and $x = \log_2((3 + \sqrt{73})/2)$. Second: $x \leq 1$.
- Q014** Use $t = 2^x > 0$ or $t = 3^x > 0$; solve the resulting quadratic and retain only roots satisfying $t > 0$.
- Q015** For each item, $t = a^x > 0$ gives a quadratic equation/inequality; convert every admissible t back to an exact logarithmic value or interval for x .
- Q016** Let $t = (1/2)^{x^2-2x} > 0$. Then $t^2 - 5t + 4 \leq 0$, so $1 \leq t \leq 4$, which gives $0 \leq x^2 - 2x \leq 2$. Hence $1 - \sqrt{3} \leq x \leq 0$ or $2 \leq x \leq 1 + \sqrt{3}$.
- Q017** Let $t = 2^x \in [1/2, 8]$. Then $y = t^2 - 4t = (t - 2)^2 - 4$: minimum -4 at $t = 2, x = 1$, maximum 32 at $t = 8, x = 3$.
- Q018** Let $t = 3^x \in [1/3, 9]$. Then $y = t^2 + 2t - 9$; the minimum is at $t = 1/3$, and the maximum is at $t = 9$.
- Q019** In each case set $t = a^x$, write the exact t -range, and compare the quadratic vertex and endpoints; report both extreme values and their x -locations.
- Q020** Let $t = 2^x \in [1/2, 4]$. Then $y = 2t^2 - 4t + 11 = 2(t - 1)^2 + 9$: minimum 9 at $x = 0$, maximum 27 at $x = 2$.
- Q021** h is non-monotonic on $\mathbb{R}$ and has global minimum $h(0) = 2$.
- Q022** C
- Q023** B
- Q024** C
- Q025** B
- Q026** B
- Q027** B
- Q028** C
- Q029** B
- Q030** A
- Q031** B
- Q032** C
- Q033** B
- Q034** C
- Q035** B
- Q036** B
- Q037** C
- Q038** B
- Q039** B
- Q040** A
- Quiz MCQ 1** A
- Quiz MCQ 2** B
- Quiz MCQ 3** C
- Quiz MCQ 4** C
- Quiz WRITTEN 5** Decreasing; $D = \mathbb{R}, R = (0, \infty), y =$
- Quiz WRITTEN 6** $2^{-3} < 2^0 < 2^{5/2}$
- Quiz WRITTEN 7** $x = 0$ or $x = 2$
- Quiz WRITTEN 8** $2atx = 0$